

electron/knowledge-graph-runner.cjs

Kind: added · Baseline: 8e0dc031dcfc · Target: NovaDiff · Generated: 2026-05-19T21:15:03.843Z

Overview

A new `electron/knowledge-graph-runner.cjs` file is added. It exposes three public APIs: `resolveRepoRoot`, `ensureGraphCoreBuilt`, and `buildKnowledgeGraph` (lines 515-671). The module replaces an earlier ad-hoc script by providing a deterministic pipeline that builds a knowledge graph for a given project root.

Key changes

- **Exports** – `resolveRepoRoot`, `ensureGraphCoreBuilt`, `buildKnowledgeGraph` (lines 515-671).
- **Node imports** – `fs`, `fsp`, `path`, `os`, `spawn`, `pathToFileURL`, `createRequire` (lines 3-9).
- **Constants** – `BATCH_SIZE`, `MAX_FILES`, `EXTRACT_TIMEOUT_MS` (lines 11-14).
- **File discovery** – `walkProjectFiles` (lines 147-185) filters directories in `SKIP_DIRS` and skips files >512 KB.
- **Language detection** – `detectLanguage` (lines 187-232) maps extensions to language IDs.
- **Import resolution** – `resolveRelativeImport` (lines 244-279) and `buildBatchImportData` (lines 281-303).
- **Graph assembly** – `mapStructuralAnalysis`, `buildFileMeta`, `buildLayersFromFiles`, `assignNodeLayers`, `buildDiffOverlay` (lines 344-496).
- **Git integration** – `getGitHash` (lines 497-504).
- **Batch extraction** – `runExtractStructureBatch` (lines 306-342) writes a temporary JSON payload, runs `graph/extract-structure.mjs`, and parses the result.
- **Progress reporting** – optional `onProgress` callback is invoked during scanning, extraction, and assembly.
- **Output** – writes `knowledge-graph.json` and, if `changedPaths` are supplied, `diff-overlay.json` into `.novadiff-graph` (lines 631-654).

Impact

- **Deterministic graph generation** via Tree-sitter; no LLM is involved.
- **Memory safety** – batch limits (`BATCH_SIZE`, `MAX_FILES`) prevent runaway usage.
- **Modular design** – functions have clear responsibilities, easing testing and maintenance.
- **Electron compatibility** – child processes are launched with `ELECTRON_RUN_AS_NODE` when needed (lines 54-58).
- **Observability** – progress callbacks provide real-time feedback during long scans.

Risks & follow-ups

- **Missing dependencies** – `packages/graph-core` or `graph/extract-structure.mjs` must exist; otherwise `ensureGraphCoreBuilt` throws (lines 521-524).

- **Spawn behavior** – verify that `spawn` with `ELECTRON_RUN_AS_NODE` works on all target platforms.
- **Large projects** – `MAX_FILES` may truncate analysis; confirm the limit is acceptable for typical repositories.
- **Error handling** – `runCommand` propagates raw stderr; callers should handle failures gracefully.